Next Word Prediction using Knowledge Distillation

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Abstract

This report presents the project "Next Word Prediction using Knowledge Distillation with Transformer-teacher and LSTM-student," aimed at enhancing next-word prediction for resource-constrained devices. By transferring knowledge from a high-capacity Transformer to a lightweight LSTM, we achieved a validation accuracy of 9.0% compared to a baseline LSTM's peak of 8.0%. Leveraging the "ArticlesMarch2018.csv" dataset, innovative preprocessing, and insights from ULMFIT, we address the challenge of efficient NLP deployment.

1 Introduction

Next-word prediction is a fundamental task in natural language processing (NLP), powering applications like predictive keyboards, virtual assistants, and text generation. It requires capturing long-range dependencies and semantic information, a challenge for resource-constrained devices such as mobile phones and IoT gadgets. Our project uses knowledge distillation, where a Transformer (teacher) transfers its contextual understanding to an LSTM (student), enabling high performance with low computational demands. Using news snippets from the "ArticlesMarch2018.csv" dataset, we compare the distilled LSTM against a baseline LSTM, achieving a validation accuracy of 9.0% versus the baseline's peak of 8.0%. This report summarizes our methodology, results, experiences, and applications.

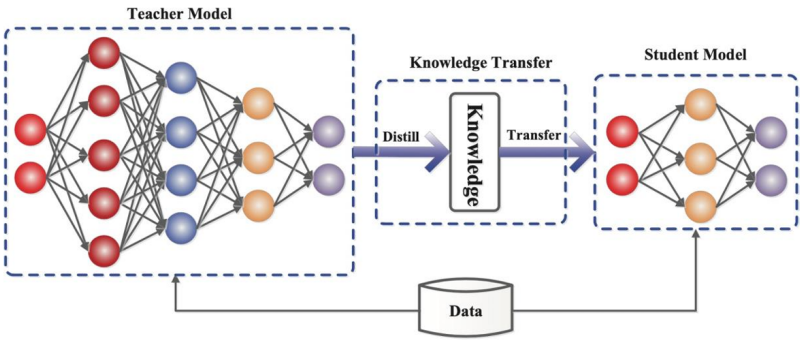

Figure 1: Proposed pipeline.

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2 Literature Review

Our project introduces "Reverse Evolution," a concept where we transfer the expressive power of Transformers back to LSTMs, aligning with our goal of efficient inference on simpler hardware.

2.1 Knowledge Distillation

Introduced by Hinton et al.(2015) in "Distilling the Knowledge in a Neural Network" (arXiv:1503.02531), knowledge distillation allows a smaller student model to learn from a larger teacher model by mimicking its soft predictions. This technique is ideal for compressing models for deployment on resource-limited devices.

2.2 Transformers

Vaswani et al. (2017) in "Attention is All you Need" (arXiv:1706.03762) established Transformers as the state-of-the-art in NLP, leveraging self-attention to capture long-range dependencies. However, their computational complexity makes them unsuitable for lightweight applications.

2.3 LSTMs

Long Short-Term Memory (LSTM) networks, a type of recurrent neural network, are efficient for sequential data and better suited for resource-constrained environments, though they may struggle with complex patterns without guidance.

2.4 ULMFIT

Howard and Ruder (2018) in "Universal Language Model Fine-tuning for Text Classification" (arXiv:1801.06146) proposed ULMFIT, a transfer learning method for NLP that pretrains a language model and fine-tunes it for specific tasks, achieving high performance with limited data. While our project uses distillation, ULMFIT's success with transfer learning informs our approach.

3 Methodology

3.1 Model Architectures

Teacher Model (Transformer): Input shape (max_len-1,), embedding layer (vocabulary size voc, dimension 200), custom TransformerBlock with 4 attention heads, feed-forward dimension 128, 4 layer normalization, and dropout (rate=0.1). Additional layers include dropout (0.2), dense (512 units, ReLU), flatten, and softmax output over the vocabulary. Compiled with Adam optimizer and categorical cross-entropy loss.

Student Model (LSTM): Input shape (max_len-1,), embedding layer (dimension 200), LSTM with 256 units, dropout (0.2), and dense output layer (logits over the vocabulary).

Baseline Model: Identical to the student, trained without distillation for comparison.

3.2 Knowledge Distillation

We implemented a custom Distiller class to train the student, using a hybrid loss: cross-entropy with true labels and KL-divergence with the teacher's soft predictions, scaled by a temperature of 2.0 and weighted by alpha of 0.5. The teacher was frozen during training, focusing the learning on the student. This approach transfers Transformer capabilities to the LSTM for efficient inference.

$$\mathcal{L}_{\text{hybrid}} = \alpha \cdot T^2 \cdot \text{KL}(p_t || p_s) + (1 - \alpha) \cdot \text{CE}(y, p_s)$$

Figure 2: Hybrid loss formula.

3.3 Training and Evaluation

Models were trained for up to 50 epochs with a validation split of 0.2, using callbacks (ReduceLROnPlateau, EarlyStopping) to prevent overfitting. Evaluation focused on validation accuracy, visualized in attached plots.

4 Experiments

4.1 Challenges:

Data Collection: The dataset "ArticlesMarch2018.csv" was sourced as a CSV file, containing news snippets. Its structure suggests a public news archive or aggregator origin, chosen for its diversity and relevance to real-world applications like mobile auto-completion.

Data Cleaning: News snippets had inconsistent formatting, requiring careful decisions on punctuation removal and lowercasing to preserve context (e.g., proper nouns).

Model Training: The Transformer was computationally intensive on Colab, leading us to test attention heads (2, 4, 8) before settling on 4 for balance.

Distillation Tuning: Initial temperature settings (e.g., 5.0) caused the student to overfit to the teacher's predictions. We iterated to a temperature of 2.0 and alpha of 0.5 for optimal balance.

4.2 Learnings:

Preprocessing is crucial for NLP tasks; our sequence generation approach, though time-consuming, was essential for capturing incremental context.

Visualization via plots helped identify overfitting and guided hyperparameter adjustments.

Knowledge distillation requires careful tuning to balance true labels and teacher guidance, a process that improved our understanding of transfer learning.

5 Results

The distilled model outperforms the baseline by 1 percentage point at peak, demonstrating that knowledge distillation enhances the LSTM's ability to capture contextual patterns. The low absolute accuracies reflect the task's difficulty—predicting the exact next word from a large vocabulary—but the relative improvement validates our approach, showing distillation's effectiveness for resource-constrained deployment.

Transformer as Teacher: Chosen for its ability to model long-range dependencies and semantic relationships, critical for news snippets' nuanced language. Implementing a custom TransformerBlock helped in tailoring the teacher model to our specific needs [Vaswani et al., 2017].

LSTM as Student: Selected for its efficiency and sequential nature, ideal for resource-constrained devices. We tested 128, 256, and 512 units, choosing 256 for balanced capacity and efficiency.

Knowledge Distillation: Used to transfer Transformer capabilities to the LSTM, aligning with "Reverse Evolution" . Tuning temperature and alpha was done to optimize knowledge transfer.

Callbacks: ReduceLROnPlateau and EarlyStopping were included to prevent overfitting, given the noisy dataset. Proper patience values (3 vs 5) helps in robust training.

Validation Accuracy: Chosen as the metric to directly measure prediction performance, with plots aiding comparison, for thorough evaluation.

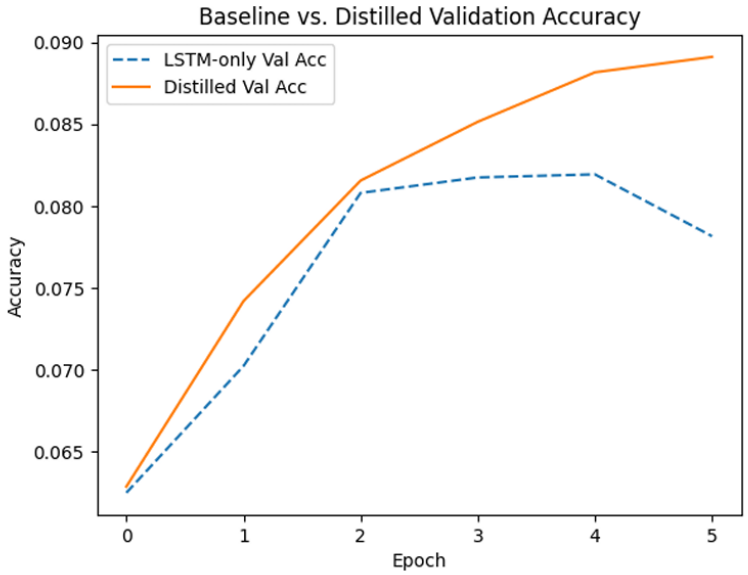

Figure 3: Comparision of validation accuracies.

6 Conclusion

Our project demonstrates the effectiveness of knowledge distillation for next-word prediction, achieving a validation accuracy of 9.0% with the distilled LSTM, compared to the baseline's peak of 8.0%. Using news snippets and innovative preprocessing, we addressed deployment challenges for resource-constrained devices, aligning with ULMFIT's transfer learning principles and our "Reverse Evolution" concept.

References

[1] Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the Knowledge in a Neural Network. arXiv:1503.02531.

[2] Vaswani, A., et al. (2017). Attention is All You Need. arXiv:1706.03762.

[3] Howard, J., & Ruder, S. (2018). Universal Language Model Fine-tuning for Text Classification. arXiv:1801.06146.